Electricity & Magnetism I: Problems on magnetic field sources

A: long straight copper wire 0.127 cm in radians carries a current of 50 amper. Find the magnetic field B at the surface of the wire ?

Solution:

B: A current of 0.5 A exists in the circuit and magnetic field in which it is immersed has a magnitude of 0.20 T. Find the length of the wire?

Solution:

C: Two straight wire A and B carrying currents of 4 and 5 amper respectively in opposite n direction. Lie parallel to each other 4 cm apart. Compute the force on a 10 cm long section of the wire B near its center

Solution :

D: Two long, straight parallel wires 8.6 cm apart carry equal current. The wires repel each other with a force per unit length of 3.6 N/m. Determine the current in each wire?

Solution:

E- A closed circuit consists of two semicircles of radii 40.0 cm and 20.0 cm that are connected by straight segments. A current of 3.00 A flows in the circuit in clockwise direction. Find the magnetic field strength at the point .

Solution: